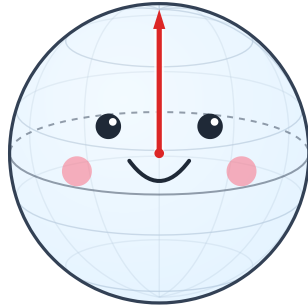


Explore the wonders of quantum computers

An observation notebook you finish while playing the puzzle game **QA²**

Today's goal

When you line up blocks, find the rules for when they clear or transform



Qubit

What you'll do

1. Play **QA²**
2. Watch how blocks (= instructions) move
3. Write the rules you notice on this sheet

What you need

☐ QA² ☐ This printout ☐ Pencil ☐ Colored pencils (optional)

Get QA²

Download free on the App Store or Google Play

[Scan a QR code with a phone](#)

iPhone / iPad



App Store

Android



Google Play

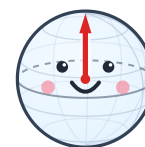
Name _____

Grade _____

School / club _____

Qubit and the secrets of the blocks

- 1 **Qubit** doesn't show up as a character inside the game, but he's the star behind the scenes. In a quantum computer, a qubit is the basic unit of data, the thing you compute with.



Qubit

- 2 Qubit's **red arrow** always points to **one place on Earth**. Sometimes Tokyo, sometimes the North Pole, sometimes the South Pole... it can point anywhere!

- 3 Now meet the **blocks**. When you give a block to Qubit, the direction he points **changes**. Blocks are the **instructions** of a quantum computer. Together with Qubit, they move the computation forward.

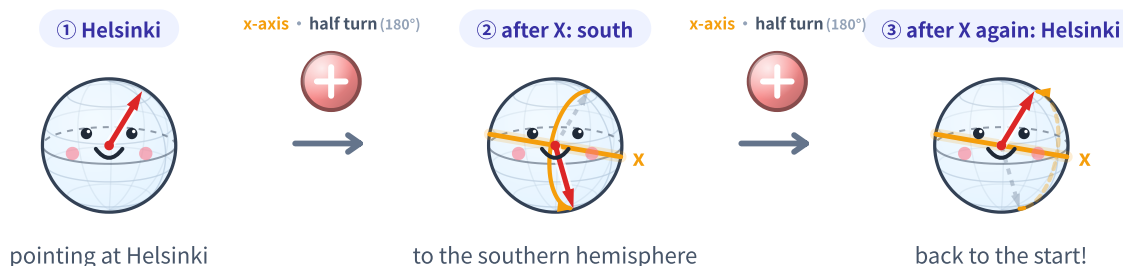


X block

- 4  is one of six block types. No matter where the arrow points, X spins it halfway around the **x-axis** (180°). First try Qubit pointing at the **North Pole**:



Now try Qubit pointing at **Helsinki, Finland**:



Different starting points, same rule: half a turn around the **x-axis**. Two X blocks make a full turn, so the arrow **returns to where it started**.

- 5 “Back to start” means giving Qubit **two**  **blocks in a row does nothing**. So if you see   stacked, you can erase them!   = **Match and clear!**

 Try it in QA²: Stack two  **blocks vertically** and check that they really disappear.

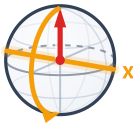
The block family (quantum gates)

When you give a block to Qubit, his **arrow** spins. Compare **which axis** it spins around and **how far**.

Half-turn blocks



X block



x-axis · half turn

Half turn around the x-axis. The mark looks like +, not the letter X.



Y block

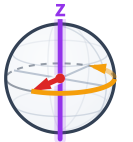


y-axis · half turn

Half turn around the y-axis. It spins a different way than .



Z block



z-axis · half turn

Half turn around the upright z-axis. North and South Poles don't move.

Diagonal half turn



H block



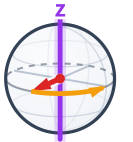
diagonal · half turn

Half turn around the diagonal axis between x and z.

Smaller turns



S block



z-axis · quarter turn

Quarter turn (90°) around z. Two S blocks transform into a  !



T block

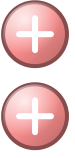
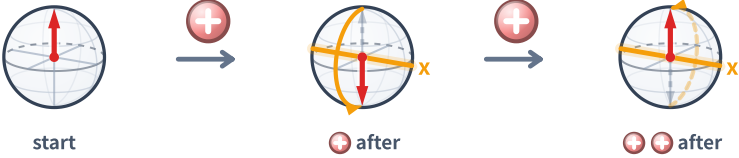
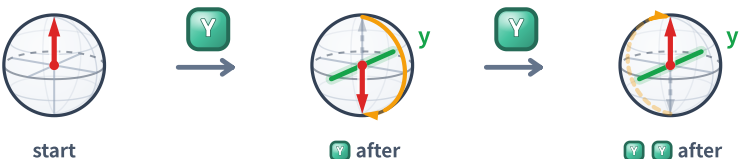
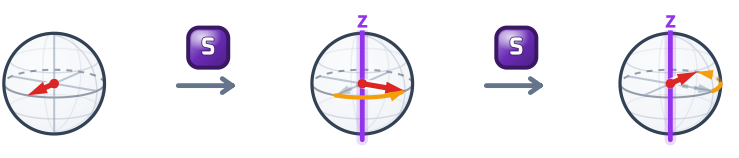
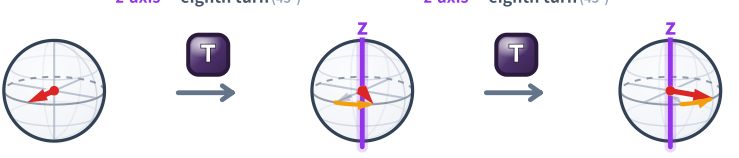


z-axis · eighth turn

Eighth turn (45°) around z. Two T blocks transform into an  !

What happens when you stack two of the same block?

How to write: Write only in the large dotted box on the left. If everything clears, write “clears”.

X² X × 2  → write here clears ↑ example	x-axis · half turn (180°) x-axis · half turn (180°)  arrow back to start → clears
Y² Y × 2  → write here ?	y-axis · half turn (180°) y-axis · half turn (180°) 
Z² Z × 2  → write here ?	z-axis · half turn (180°) z-axis · half turn (180°) 
H² H × 2  → write here ?	diagonal · half turn (180°) diagonal · half turn (180°) 
S² S × 2  → write here ?	z-axis · quarter turn (90°) z-axis · quarter turn (90°) 
T² T × 2  → write here ?	z-axis · eighth turn (45°) z-axis · eighth turn (45°) 

Sandwich with H — the middle one transforms

The outer two H blocks clear, and the block in the middle **turns into a different block**. Think about it the same way as on the previous page.

HXH

H · X · H

H

+

H

→

write here

Z

↑ example

diagonal · half turn (180°)

start

→

H

→

diagon

→

+

→

diagonal · half turn (180°)

→

H

→

diagon

after

after

after

half turn around z = same as Z

HZH

H · Z · H

H

Z

H

→

write here

?

diagonal · half turn (180°)

start

→

H

→

diagon

→

Z

→

diagonal · half turn (180°)

→

H

→

diagon

after

after

after

HYH

H · Y · H

H

Y

H

→

write here

?

diagonal · half turn (180°)

start

→

H

→

diagon

→

Y

→

diagonal · half turn (180°)

→

H

→

diagon

after

after

after

Predict / notice

When you sandwich with H , + becomes _____

When you sandwich with H , Z becomes _____

When you sandwich with H , Y becomes _____

Other things I noticed:

Sandwich with S or T?

Same idea: put the same block on both outsides and watch. Sometimes everything clears. Think about it the same way as on the previous page.

SXS $S \cdot X \cdot S$

S

+

S

→

+

 ↑ example

start

S

 →

S after

+

 →

S + after

S

 →

S + S after

z-axis · quarter turn (90°) x-axis · half turn (180°) z-axis · quarter turn (90°)

half turn around x = same as +

SYS $S \cdot Y \cdot S$

S

Y

S

→

write here
 ?

start

S

 →

S after

Y

 →

S Y after

S

 →

S Y S after

z-axis · quarter turn (90°) y-axis · half turn (180°) z-axis · quarter turn (90°)

SZS $S \cdot Z \cdot S$

S

Z

S

→

write here
 ?

start

S

 →

S after

Z

 →

S Z after

S

 →

S Z S after

z-axis · quarter turn (90°) z-axis · half turn (180°) z-axis · quarter turn (90°)

Try sandwiching with T

TST $T \cdot S \cdot T$

T

S

T

→

write here
 ?

start

T

 →

T after

S

 →

T S after

T

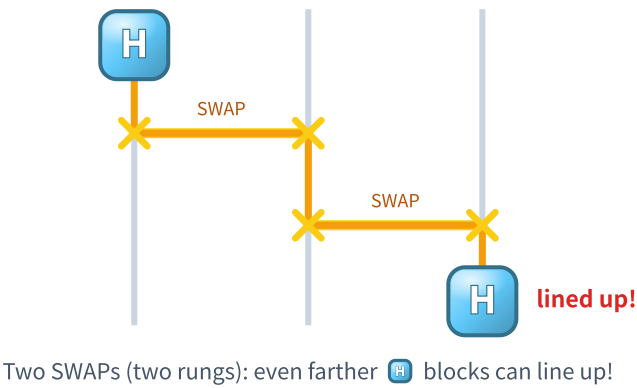
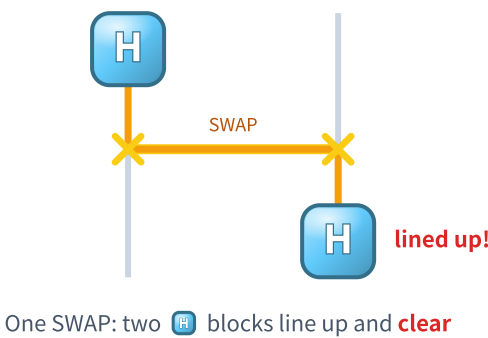
 →

T S T after

z-axis · eighth turn (45°) z-axis · quarter turn (90°) z-axis · eighth turn (45°)

Bonus: SWAP is an instruction that swaps two lanes

SWAP swaps two paths (lanes) connected by a bar. Follow the wires like a zigzag maze. Blocks that looked far apart can end up **stacked**!



Challenge: Build a long SWAP “ladder,” match through it, and score big in QA². Try different wirings and line up the clearing matches you already know, across SWAP.

What I found

Blocks that lined up: _____

Number of SWAPs I used: _____

What I noticed:

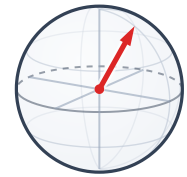


Name _____ Date _____

Background for parents & teachers

Qubits and our character “Qubit”

In a normal computer, the smallest piece of information, a **bit**, is only **0** or **1**. A quantum computer’s **qubit** is like our character Qubit: it can point in **any direction on a sphere** (the Bloch sphere). North Pole $\leftrightarrow$ $|0\rangle$, South Pole $\leftrightarrow$ $|1\rangle$. In between are **superpositions**, states a classical bit can’t hold. That’s what unlocks new kinds of computation.



North Pole = $|0\rangle$ · South Pole = $|1\rangle$
in between = superposition

Quantum gates = the blocks



The instructions that change a qubit are **quantum gates**, exactly the blocks in this worksheet. The symbols **H, X, Y, Z, S, T** are the same ones used in university textbooks and research papers. Through the game, kids meet real quantum-computing notation.

In math terms (optional depth)

A qubit state is a **state vector**.

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \text{North Pole} \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \text{South Pole} \quad |+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{Equator}$$

Gates are **unitary matrices**, rotations around an axis.

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{x-axis } 180^\circ \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \text{z-axis } 180^\circ \quad S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \quad \text{z-axis } 90^\circ \quad H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \text{x+z axis } 180^\circ$$

“Handing Qubit a block” so the red arrow moves (pp. 2–6) is what matrix $\times$ vector looks like as a picture.

$$\text{例1: } X|0\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

→ North Pole flips to South Pole (p.2).

$$\text{例2: } X \cdot X|0\rangle = X|1\rangle = |0\rangle, \quad X \cdot X|1\rangle = X|0\rangle = |1\rangle, \quad \text{so } X \cdot X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \quad \text{identity}$$

→ Two X restore the start; “match and clear” means the stack equals the identity (p.4 X^2).

$$\text{例3: } S \cdot S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = Z$$

→ Two S act like Z (p.4); T is 45° on z (phase $e^{i\pi/4}$); two T make S.

Why this worksheet

Through play and careful watching, learners meet core ideas (superposition, quantum gates, unitarity) using the real symbols. Finding rules themselves (XX clears, $S^2 = Z$, and so on) builds a foundation for later study. Suggested uses: STEM club session, science-fair project log, classroom lab (1–2 periods), or a weekend project at home.